CO3 - AT3 -SCENARIO BASED QUESTONS

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1. Multilayer Perceptron & Backpropagation Algorithm (5 Marks)

Backpropagation Overview

Backpropagation (Backward Propagation of Errors) is a supervised learning algorithm used to train neural networks. It computes the gradient of the loss function with respect to each weight using the **chain rule of calculus**, moving backward from the output layer to the input layer to adjust the weights and minimize prediction error.

Step-by-Step Mechanism

A. Forward Pass

1. Input x is fed into the network.
2. Weighted sum (net input) is calculated for hidden neuron h:

   zh=w1x1+w2x2+bh

3. Activation function σ (e.g., Sigmoid) is applied to produce output ah:

   ah=σ(zh)=1+e−zh1

4. The process repeats up to the final output layer to produce predicted value y^.

B. Error Calculation

For binary classification, Binary Cross-Entropy or Mean Squared Error (MSE) is used. Using MSE for simplicity:

E=21(y−y^)2

(where y is the actual target and y^ is the predicted output).

C. Backward Pass (Weight Updates)

Using the chain rule, the gradient of error with respect to an output weight who

is: ∂who∂E=∂y^∂E⋅∂zo∂y^⋅∂who∂zo

Where:

- ∂y^∂E=−(y−y^)
- ∂zo∂y^=y^(1−y^) *(Derivative of Sigmoid)*

- $∂who∂zo=ah$

The weight update rule using learning rate η:

$$who(new)=who-\eta\cdot∂who∂E$$

Conceptual Concrete Example

Consider a single-neuron network:

- **Inputs:** $x=1$
- **Initial Weight:** $w=0.5$, **Bias:** $b=0$, **Learning Rate:** $\eta=0.1$, **Target:** $y=1$ 1.

Forward Pass:

$$z=(1\times0.5)+0=0.5$$

$$y\hat{}=\sigma(0.5)=1+e-0.51\approx0.622$$

2. **Error Calculation:**

$$E=21(1-0.622)2=0.0714$$

3. **Backward Pass:**

$$\delta=-(y-y\hat{})\cdot y\hat{}(1-y\hat{})=-(1-0.622)\cdot(0.622)(1-0.622)=-0.378\cdot0.235\approx-0.0888$$

$$∂w∂E=\delta\cdot x=-0.0888\times1=-0.0888$$

4. **Weight Update:**

$$w(new)=0.5-0.1(-0.0888)=0.50888$$

Influence of Hyperparameters

- **Learning Rate (η):**
  - **Too high ($\eta>1.0$):** Causes the model to overshoot local minima, leading to divergence or unstable oscillations.
  - **Too low ($\eta<0.001$):** Traps the model in local minima or results in extremely slow convergence.
- **Activation Functions:**
  - **Sigmoid / Tanh:** Subject to the **vanishing gradient problem** for deep networks because derivatives approach zero for extreme inputs.
  - **ReLU (Rectified Linear Unit):** Avoids vanishing gradients, allowing faster convergence, though it can suffer from "dying ReLU" if learning rates are uncalibrated.

Question 2: Genetic Algorithm Optimization (5 Marks)

Problem Setup & Chromosome Mapping

We represent candidate health profiles from the dataset as **binary chromosomes** to optimize a **Disease Risk Fitness Function**.

Encoding Schema:

Each chromosome contains 4 feature genes:

1. **Age Gene (2 bits):** `30` $\rightarrow$ `00`, `40` $\rightarrow$ `01`, `45` $\rightarrow$ `10`, `50/60` $\rightarrow$ `11`
2. **BP Gene (1 bit):** `Normal` $\rightarrow$ `0`, `High` $\rightarrow$ `1`
3. **Glucose Gene (1 bit):** `Normal` $\rightarrow$ `0`, `High` $\rightarrow$ `1` 4. **Lifestyle Risk Gene (2 bits):** `Low` $\rightarrow$ `00`, `Medium` $\rightarrow$ `01`, `High` $\rightarrow$ `10`

Fitness Function:

$$\text{Fitness} = 0.2 \times \text{Age Score} + 0.3 \times \text{BP Score} + 0.3 \times \text{Glucose Score} + 0.2 \times \text{Lifestyle Score}$$

*(Scores scaled to 0–100; Fitness ≥ 60 indicates a **Suitable / High Risk (Yes)** candidate solution).*

Initial Population (Mapped directly from Dataset Examples 1 to 4)

ID	Data Example
P1	Ex 1 (45, High, High, High)
P2	Ex 2 (50, Normal, High, Medium)
P3	Ex 3 (30, Normal, Normal, Low)

Bi Ch e
`10`

`11 0 1 01`
`00 0 0 00`

Suitable?

Yes	ID	Data Example
Yes	P4	Ex 4 (60, High, High, High)
No		

11 1 1 10	

Suitable?
Yes

Evolutionary Operations 1.

Selection:

Bi Cl e

- **Roulette Wheel / Fitness Proportional Selection:** Fitter individuals have a higher selection probability. Here, **P4** (95.0) and **P1** (85.0) dominate the selection pool, while **P3** (0.0) is eliminated due to zero suitability.

2. **Crossover:**
   - Recombines parent genes to produce higher-performing offspring.
   - Single-point crossover between **P1** (`1011|10`) and **P4** (`1111|10`) at locus 4: ▪ **Offspring 1 (O1):** `1011 + 10` $\rightarrow$ `101110` (Fitness = **85.0**) ▪ **Offspring 2 (O2):** `1111 + 10` $\rightarrow$ `111110` (Fitness = **95.0**) 3. **Mutation:**
   - Flips a bit to explore unvisited hypothesis spaces and avoid local minima. ○ **Mutation on O2:** Flipping the 1st bit (`1` $\rightarrow$ `1`) keeps max age, and maintaining high values across all fields results in max risk representation `11 1 1 11` (Risk = Max).

Population Evolution Across Generations

Generation 0 (Initial)

- Average Fitness = $\frac{85 + 60 + 0 + 95}{4} = \mathbf{60.0}$

Generation 1 (Post Crossover)

- Candidates: **O1** (`101110`), **O2** (`111110`), **P4** (`111110`), **P1** (`101110`) •

Average Fitness = $\frac{85 + 95 + 95 + 85}{4} = \mathbf{90.0}$

Generation 2 (Post Targeted Mutation & Final Convergence)

Solution

Candidate ID	Bina Chro	Evalua		Suitable?
G2_1 (Best)	11 1	Age: 6C Glucos High		Yes
G2_2	11 1	Age: 6C Glucos Med		Yes
G2_3	10 1	Age: 45 Glucos High		Yes
G2_4	11 0	Age: 6C Glucos High		Yes

Final Optimized

· **Optimal Chromosome:** `111110` (Corresponding to Dataset Example 4) · **Optimal Feature Combination:** Age = 60, BP = High, Glucose Level = High, Lifestyle Risk = High

· **Maximum Achieved Fitness: 95.0 (Suitable: Yes)**

Question 3: Neural Networks for Disease Classification & Evaluation (5 Marks)

Backpropagation Training Process on the Dataset

1. **Feature Numerical Encoding:**
   - `Blood Pressure`: Normal = 0, High = 1
   - `Glucose Level`: Normal = 0, High = 1
   - `Lifestyle Risk`: Low = 0, Medium = 1, High = 2
   - `Actual / Predicted Output`: No = 0, Yes = 1
2. **Forward Pass:** The features pass through the input layer into a hidden layer where weighted sums ($z = \sum w_i x_i + b$) are transformed via activation functions (e.g., Sigmoid) to output a probability prediction $\hat{y}$.
3. **Loss Computation:** Binary Cross-Entropy evaluates error between target y and prediction $\hat{y}$.
4. **Backward Pass:** Gradients are propagated backward using the chain rule to update weights ($w \leftarrow w - \eta \frac{\partial E}{\partial w}$), iteratively refining boundary predictions.

Confusion Matrix Evaluation

Taking **Disease Present = Yes (1)** as Positive and **Disease Absent = No (0)** as Negative:

- **True Positives (TP):** Actual = Yes, Predicted = Yes $\rightarrow$ **Examples 1, 4** $\implies \mathbf{TP = 2}$
- **True Negatives (TN):** Actual = No, Predicted = No $\rightarrow$ **Example 3** $\implies \mathbf{TN = 1}$
- **False Positives (FP):** Actual = No, Predicted = Yes $\rightarrow$ **Example 5** $\implies \mathbf{FP = 1}$
- **False Negatives (FN):** Actual = Yes, Predicted = No $\rightarrow$ **Example 2** $\implies \mathbf{FN = 1}$

Performance Metrics Calculation

1. **Accuracy:**

   $$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{Total Samples}} = \frac{2 + 1}{5} = \frac{3}{5} = \mathbf{60.0\%}$$

2. **Precision:**

   $$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} = \frac{2}{2 + 1} = \frac{2}{3} \approx \mathbf{66.67\%}$$

3. **Recall (Sensitivity):**

   $$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} = \frac{2}{2 + 1} = \frac{2}{3} \approx \mathbf{66.67\%}$$

4. **F1-Score:**

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = 2 \times \frac{0.6667 \times 0.6667}{0.6667 + 0.6667} = \mathbf{66.67\%}$$

Impact of Preprocessing & Feature Selection

1. **Data Preprocessing:**
   - **Feature Normalization / Scaling:** In this dataset, `Age` ranges from 30 to 60, whereas encoded values for `Blood Pressure` range between 0 and 1. Without scaling (e.g., Min-Max or Z-score normalization), gradients for `Age` will dominate weight updates, leading to skewed gradient descent paths and slow convergence.
   - **One-Hot Encoding:** Converting categorical variables (`Lifestyle Risk`: Low, Medium, High) into one-hot binary vectors ensures the model does not infer incorrect numerical distances between categories.
2. **Feature Selection:**
   - Identifies high-impact diagnostic markers (such as `Glucose Level` and `Blood Pressure`) while dropping redundant or noisy attributes.
   - Reduces dataset dimensionality, preventing **overfitting** on small sample sizes (like this 5-example set) and enhancing overall model generalization.